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Macro-Systematic Approach to Multi-Asset Investing

Version 1.0

Alpha Capital Jr.
Buy-Side

Macro Regime-Based Approach to Multi-Asset Investing

Systematic Framework for Dynamic Portfolio Allocation Across Economic Cycles

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1 Foreword

1.1 Presentation

My name is Nicolas Poncin, and I am a Master's student in Financial Engineering.

This project is developed within **Alpha Capital Jr.**, a student-led initiative co-founded with Basile Houriez. The initiative brings together four finance enthusiasts working on complementary research projects in quantitative and financial markets. It provides a collaborative environment in which students can combine complementary skills, exchange perspectives, and critically challenge one another's ideas.

The development of this framework is documented and regularly shared through Alpha Capital Jr.'s LinkedIn page and my personal LinkedIn profile. This public documentation aims to ensure transparency, facilitate feedback, and track the progressive development of the project.

1.2 Problem Statement

Traditional multi-asset portfolios are often built on relatively stable assumptions regarding expected returns, volatility, and cross-asset diversification. However, these assumptions may become unreliable as financial and macroeconomic conditions evolve.

In particular:

- Asset return correlations are time-varying and may change substantially across market environments;
- Expected risk premia and volatility dynamics depend on the prevailing macroeconomic regime;
- Diversification benefits may weaken or disappear during periods of market stress, when correlations between risky assets tend to increase.

Consequently, static asset allocations may be poorly suited to environments characterised by changing growth, inflation, monetary policy, and market-risk conditions. A portfolio that appears well diversified during normal periods may become significantly more concentrated in its underlying risk exposures during crises.

1.3 Objective

This motivates the development of a systematic, macro-driven allocation framework capable of identifying different economic regimes and adapting assets exposures accordingly.

The objective is not to forecast with certainty , but to improve the portfolio's ability to manage regime-dependent risk while preserving exposure to long-term risk premia.

2 Framework

2.1 Regime Model

The framework reduces macroeconomic variables into a structured and decision-oriented investment system. It is built on the premise that asset price behavior is primarily driven by two dominant macro forces:

- **Economic Growth**
- **Monetary Policy**

Rather than forecasting asset returns directly, the objective is to summarize the macro environment into a small number of interpretable signals that can be mapped into investable portfolio regimes.

2.2 Variables

The first axis captures the **state of the economy**. In the current version of the model, it is represented by three variables:

2.2.1 Unemployment

The unemployment rate is denoted by u. A rising unemployment rate is generally associated with an economic slowdown. The model uses the U.S. unemployment rate, identified by the FRED series `UNRATE`, expressed as a percentage of the labour force.

The series is monthly and seasonally adjusted. Since the objective is not to forecast the unemployment rate itself, but rather to use it as an indicator of the current economic environment, the model uses its level as a measure of the state of the labour market.

2.2.2 Economic growth or activity

Economic growth is denoted by g. In the current version of the framework, U.S. economic activity is proxied by the Industrial Production Total Index, identified by the FRED series `INDPRO`.

The original series is monthly, seasonally adjusted, and expressed as an index with a base equal to 100 in 2017. Since the level of an index is not directly comparable over time and may contain persistent trends, it is transformed into a year-on-year percentage change. No outlier treatment is applied to this variable. Extreme observations are deliberately retained in order to preserve unusual macroeconomic conditions that may contain relevant information for regime detection.

2.2.3 Corporate earnings

Corporate earnings are denoted by *earnings* or e. The model uses Corporate Profits After Tax without Inventory Valuation Adjustment and Capital Consumption Adjustment, identified by the FRED series `CP`.

The original data are quarterly, seasonally adjusted, and expressed in billions of dollars at a seasonally adjusted annual rate. The series is deflated using the inflation measure described below and is subsequently transformed into a year-on-year percentage change. No outlier treatment is applied to corporate earnings. As with industrial production, the objective is to retain unusual observations that may provide information about the prevailing macroeconomic regime.

The second axis captures monetary policy conditions and is proxied by two variables:

2.2.4 Interest rates

Interest rates are denoted by *IR*. The model uses the yield on two-year U.S. Treasury securities, identified by the FRED series `DGS2`. This series measures the market yield on U.S. Treasury securities at a two-year constant maturity, quoted on an investment basis.

The original data are daily, expressed as percentages, and are not seasonally adjusted. Because the series does not contain observations for non-trading days, missing observations are approximated using the average of the preceding and following available observations.

The interest-rate series displays substantial changes across historical monetary-policy regimes. In particular, interest-rate levels observed before 2000 are generally much higher than those observed during the post-2008 period, which included quantitative easing and a prolonged period of near-zero interest rates. Consequently, the reference period used to estimate the mean and standard deviation of the interest-rate z-score (See Section 2.3.1) is restricted to the period beginning in 2008. This choice is intended to make the standardization more representative of the modern monetary-policy environment rather than of earlier high-rate regimes.

2.2.5 Inflation

Inflation is denoted by *pi or i*. The model uses the Consumer Price Index for All Urban Consumers, All Items, in U.S. City Average, identified by the FRED series `CPIAUCSL`.

The original series is monthly and seasonally adjusted. Inflation is measured as the year-on-year percentage change in the Consumer Price Index.

2.3 Variable Construction

Each macro variable is transformed into two complementary components:

1. a **level score**, which measures where the current observation stands relative to history;
2. a **momentum score**, which measures the recent direction of change in standardized form.

2.3.1 Level score

For a generic variable x_t, the standardized level score is defined as:

$$z_t(x) = \frac{x_t - \mu_x}{\sigma_x} \tag{1}$$

Where μ_x and σ_x denote the historical mean and standard deviation computed over the reference sample.

2.3.2 Momentum score

The momentum score complements it by measuring the variable's recent direction of change. For example, an unemployment rate of 7% may generally be interpreted as a negative signal, but if unemployment was 10% a few quarters earlier, the decline may indicate an improving labour market and an economic recovery.

For each variable, momentum is computed as the change between the current observation and the observation corresponding to a selected lag. The lag is expressed in the appropriate number of observations for daily, monthly, or quarterly data. The resulting signal is standardized using the mean and population standard deviation estimated from non-missing observations since 1 January 1985.

To limit the influence of extreme observations, the standardized momentum signal is clipped to the interval $[-2.5, 2.5]$. The resulting score preserves the direction and relative strength of recent changes while preventing individual observations from dominating the composite indicator. The formal definitions are provided in Appendix 6.3.

2.3.3 Variable Score

The final score for each variable is then defined as the average of the two components:

$$s_t(x) = \frac{z_t(x) + m_t(x)}{2} \tag{2}$$

To preserve a consistent economic interpretation across variables, some indicators are sign-adjusted. In particular, unemployment, interest rates, and inflation are multiplied by -1, so that higher values of the final score always correspond to a more supportive macroeconomic backdrop.

2.4 Composite Scores

The variable-level scores are then aggregated into two composite indicators.

The **economic score** is defined as:

$$\text{Econ}_t = \frac{u_t^* + g_t^* + e_t^*}{3} \tag{3}$$

where u_t^*, g_t^*, and e_t^* denote the final scores associated with unemployment, growth, and earnings:

The **monetary policy** score is defined as:

$$\text{Policy}_t = \frac{r_t^* + \pi_t^*}{2} \tag{4}$$

where IR_t^* and π_t^* denote the final scores associated with interest rates and inflation.

At this stage, all variables receive equal weights. This choice keeps the model transparent and easy to interpret.

A future extension of the framework could estimate differentiated coefficients in order to reflect the relative informational content of each variable more accurately.

2.5 Regime Matrix

The pair $Econ_t$, $Policy_t$ determines the macroeconomic regime at time t. Using a zero threshold, the framework defines four regimes:

$$\text{Regime 1: } \text{Econ}_t > 0 \quad \text{and} \quad \text{Policy}_t > 0 \tag{5}$$

$$\text{Regime 2: } \text{Econ}_t > 0 \quad \text{and} \quad \text{Policy}_t \leq 0 \tag{6}$$

$$\text{Regime 3: } \text{Econ}_t \leq 0 \quad \text{and} \quad \text{Policy}_t > 0 \tag{7}$$

$$\text{Regime 4: } \text{Econ}_t \leq 0 \quad \text{and} \quad \text{Policy}_t \leq 0 \tag{8}$$

These four quadrants provide a simple and interpretable regime map linking macroeconomic conditions to expected cross-asset behaviour. They form the basis for the subsequent analysis of asset performance and portfolio allocation across regimes.

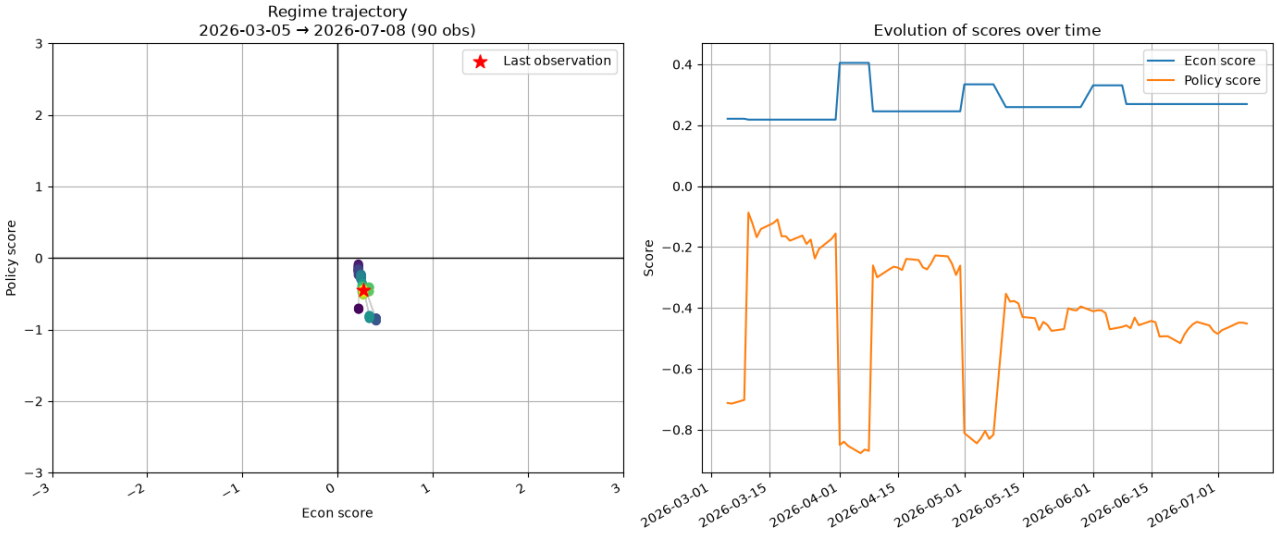


Figure 1: Evolution of regime of the last 90 observations (from 08-07-26)

3 Asset universe and allocation

3.1 Asset universe

Class	Product	Ticker
Equity	S&P 500 ETF	SPY
Commodities	Gold	GLD
Commodities	Commodity Index Fund	DBC
Bonds	Treasuries 20Y+	TLT

3.2 Assets performance in regimes

As an initial validation of the regime framework, each observation in a In Sample dataset, running from 2006 to Dec 2019 , was assigned to one of the four macroeconomic regimes:

$$S_t \in \{1, 2, 3, 4\}, \tag{9}$$

where the regimes are defined by the signs of the Economic and Policy scores:

$$\begin{aligned}
S_t = 1: &\quad \text{Economic } +, \text{ Policy } + \quad \textit{incl.}1060\ \textit{observations} \\
S_t = 2: &\quad \text{Economic } +, \text{ Policy } - \quad \textit{incl.}443\ \textit{observations} \\
S_t = 3: &\quad \text{Economic } -, \text{ Policy } + \quad \textit{incl.}1823\ \textit{observations} \\
S_t = 4: &\quad \text{Economic } -, \text{ Policy } - \quad \textit{incl.}172\ \textit{observations}.
\end{aligned}$$

For each asset and each regime, performance was then evaluated using only the observations belonging to that regime. This procedure provides a first empirical assessment of whether the assets display distinct performance profiles across macroeconomic environments.

3.2.1 Assets performance in regime 1

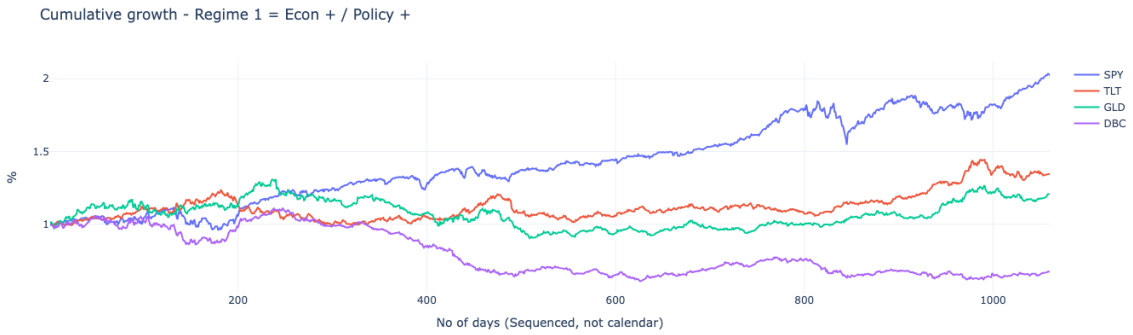


Figure 2: Asset performance in regime 1

SPY is the strongest asset in this regime, with a CAGR of 18.22%, the highest Sharpe ratio 1.317, the lowest maximum drawdown -16.12%, and the highest hit rate 57.64%.

TLT also performs positively, with a CAGR of 7.22% and a Sharpe ratio of 0.646, but its maximum drawdown is larger at -19.78%. DBC and GLD produce a lower CAGR and a Sharpe ratio.

Although its annualized volatility is relatively high, the combination of return and risk-adjusted performance supports a full allocation to SPY.

Asset	N_t	CAGR	Vol.	Sharpe	MaxDD	Hit Rate
DBC	1060	-8.94%	15.11%	-0.544	-45.13%	49.63%
GLD	1060	4.60%	14.38%	0.385	-31.04%	51.23%
SPY	1060	18.22%	13.39%	1.317	-16.12%	57.64%
TLT	1060	7.22%	11.88%	0.646	-19.78%	53.39%

Table 1: Asset performance conditional on Regime 1: Economic +, Policy +

3.2.2 Assets performance in regime 2

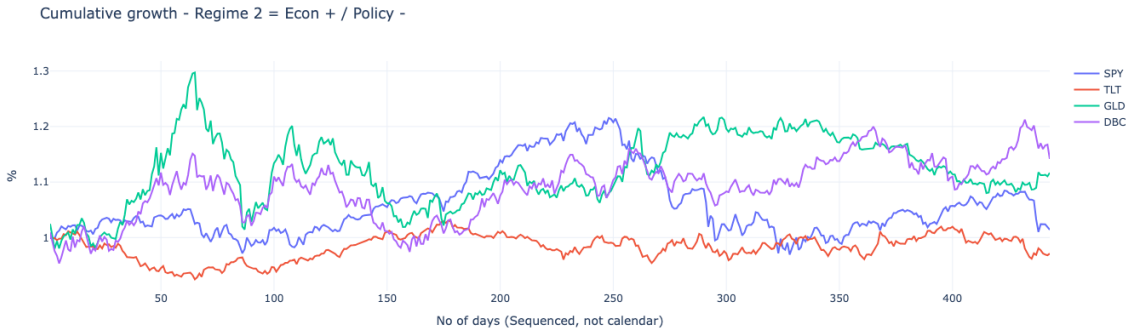


Figure 3: Asset performance in regime 2

In Regime 2, DBC and GLD are the best performers with CAGRs of 7.83% ans 6.58% respectively. But GLD is more volatile: 18.98% vs 16.78%. For this reason DBC presents a better Sharpe 0.533, despite being relatively low. TLT is the weakest asset, recording a negative CAGR and Sharpe ratio, but the smaller maximum drawdown.

Even though DBC is the best performer, and so, the chosen asset in regime 2, the evidence for a unique preferred asset is weaker in this regime, which justifies treating Regime 2 as less conclusive.

Asset	N_t	CAGR	Vol.	Sharpe	MaxDD	Hit Rate
DBC	443	7.83%	16.78%	0.533	-15.37%	49.66%
GLD	443	6.48%	18.98%	0.426	-21.79%	53.05%
SPY	443	0.79%	13.02%	0.126	-20.25%	53.05%
TLT	443	-1.65%	8.56%	-0.151	-8.57%	48.53%

Table 2: Asset performance conditional on Regime 2: Economic +, Policy −

3.2.3 Assets performance in regime 3

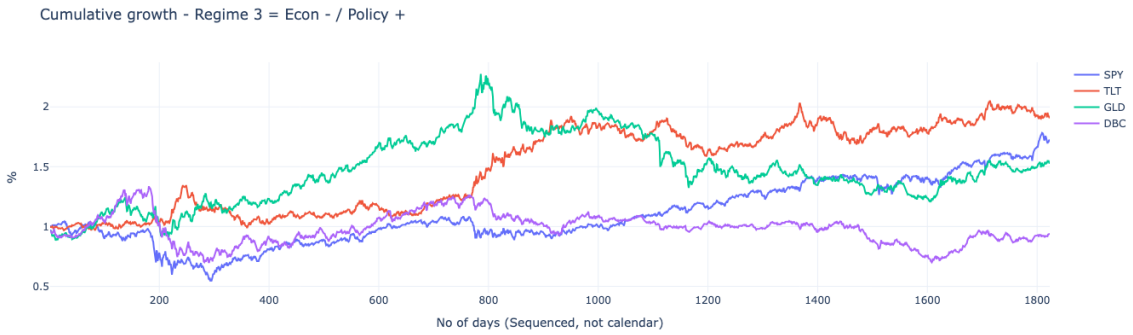


Figure 4: Asset performance in regime 3

TLT provides the most attractive risk-adjusted profile in this regime. It achieves the highest CAGR among the four assets at 9.41%, the highest Sharpe ratio at 0.646, and an annualized volatility of 15.89%. Its maximum drawdown is also comparatively limited at -26.59%. while, every other assets showcase very big maximum drawdowns (close to 50%) which supports TLT as the preferred allocation in this regime.

Asset	N_t	CAGR	Vol.	Sharpe	MaxDD	Hit Rate
DBC	1823	-0.85%	20.73%	0.062	-47.87%	50.69%
GLD	1823	6.07%	19.44%	0.401	-47.11%	52.55%
SPY	1823	7.78%	22.11%	0.449	-47.95%	54.20%
TLT	1823	9.41%	15.89%	0.646	-26.59%	53.98%

Table 3: Asset performance conditional on Regime 3: Economic −, Policy +

3.2.4 Assets performance in regime 4

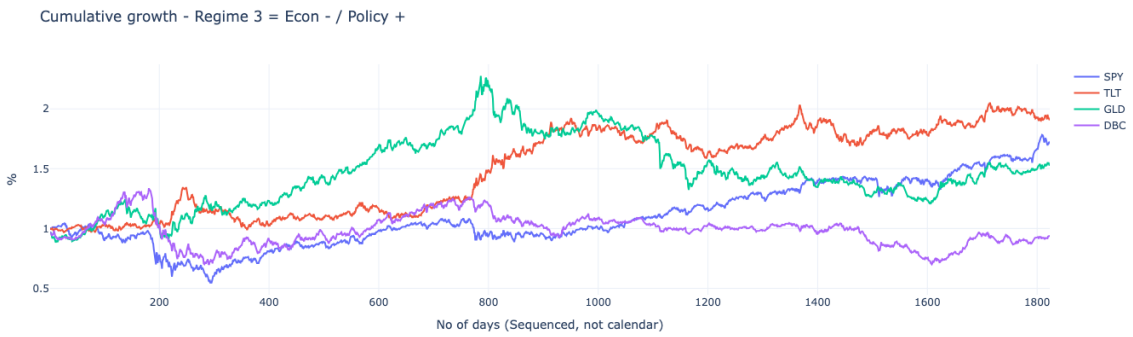


Figure 5: Asset performance in regime 4

GLD is the strongest asset in Regime 4, with a CAGR of 33.29%, a Sharpe ratio of 1.187, and the highest hit rate at 55.81%. But it showcases a maximum drawdown of -28.37%, despite and relatively high annualized volatility of 27.30%. SPY and TLT both generate negative CAGRs. Furthermore, TLT also has a low volatility, which makes it a potential candidate for shorting during regime 4. (tbi in future versions). GLD is the preferred asset in this regime.

Asset	N_t	CAGR	Vol.	Sharpe	MaxDD	Hit Rate
DBC	172	0.91%	27.54%	0.170	-28.37%	51.16%
GLD	172	33.29%	27.30%	1.187	-23.80%	55.81%
SPY	172	-6.53%	22.07%	-0.195	-17.73%	54.65%
TLT	172	-7.04%	13.33%	-0.481	-14.10%	49.42%

Table 4: Asset performance conditional on Regime 4: Economic −, Policy −

3.2.5 Summary of assets performance in regimes

	Policy ↑	**Policy ↓**
Econ ←	**R1** Equities ↑	**R2** Commodities ↑ Gold ∼
Econ →	**R3** Bonds ↑ Equities ∼	**R4** Gold ↑

Table 5: Summary of asset performance conditional on Regimes

3.3 Allocation

As a first and intentionally simplified portfolio-construction exercise, the final allocation selects the asset with the strongest observed performance within each regime. The portfolio is then fully invested in that asset. The resulting allocation rule is:

$$\mathcal{A}(S_t) = \begin{cases} 100\% \text{ SPY}, & S_t = 1, \\ 100\% \text{ DBC}, & S_t = 2, \\ 100\% \text{ TLT}, & S_t = 3, \\ 100\% \text{ GLD}, & S_t = 4. \end{cases} \tag{10}$$

The economic interpretation is straightforward. SPY is selected in the favourable growth Regime 1, TLT in the weak-growth and supportive-policy regime 3, and GLD in the adverse growth and policy Regime 4. DBC is selected in the positive-growth but restrictive-policy Regime 2 as an exposure to commodity.

4 Portfolio Backtest

4.1 Rules

- Long only portfolio
- No cash, 100% is allocated
- No stop loss, no drawdown control, no leverage
- Daily price = close price
- if regime changes at t, the allocation is changed at $t + 1$
- Risk free equals to SHV (Money Market fund) average daily returns
- In sample dataset runs from 2006 to 2019, Out of sample dataset runs from 2020 and onward.

4.2 Results (Out of sample)

This final backtest evaluates the portfolio generated by the regime-based allocation rule over the **out of sample dataset (2020 and onward)**. The allocation is therefore determined using the regime observed at date t, while the portfolio return is measured over the subsequent period $t + 1$. This timing convention ensures that the return calculation does not use information from the future.

Metric	Strategy	SPY
CAGR	14.23%	14.98%
Annualized volatility	18.60%	19.96%
Sharpe ratio	0.7222	0.7204
Sortino ratio	0.9315	1.0173
Maximum drawdown	-27.79%	tbd
Annualized turnover	28.6698	0

Table 6: Strategy vs SPY - OOS

The out-of-sample results show that the regime-based strategy does not outperform a passive investment in SPY in terms of absolute return. Its CAGR is 14.23% vs 14.98% for SPY, and its lower annualized volatility leads to a marginally higher Sharpe ratio of 0.7222 vs 0.7204. However, the strategy has a lower Sortino ratio and generates substantial annualized turnover, indicating that its smaller sharpe comes from frequent reallocations rather than superior downside efficiency. The path of cumulative performance also reveals an important weakness *(figure 6)*. The strategy performs relatively well until mid-2022, but subsequently enters a prolonged drawdown and stagnation phase, while SPY recovers and outperforms. Therefore, it provides some diversification and volatility reduction, but fails to deliver persistent out-of-sample alpha.

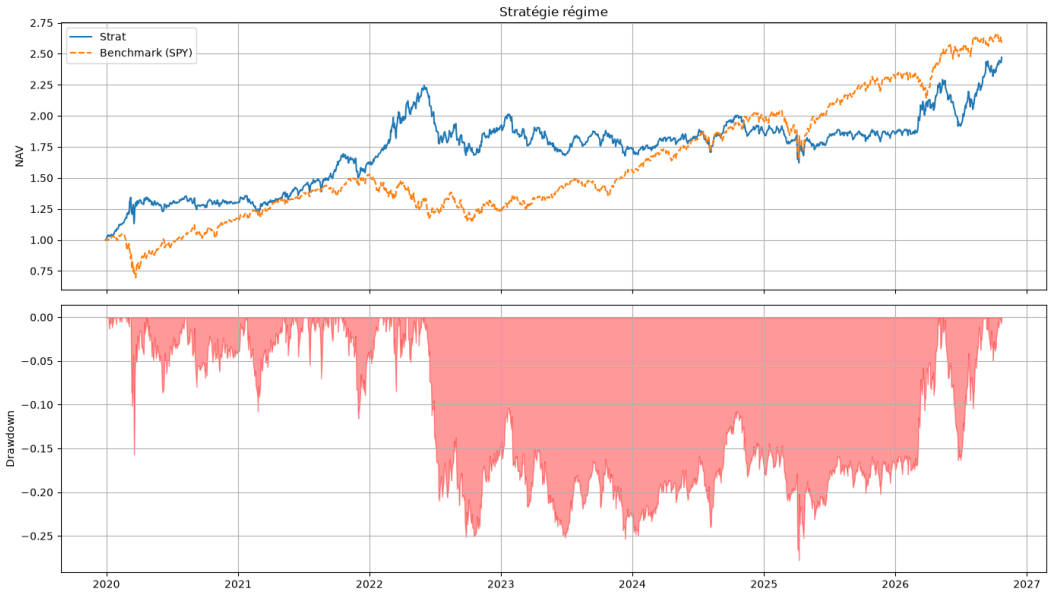


Figure 6: Performance comparison between strategy and SPY over OOS period

4.3 Historical Results and Comparison with S&P500 (No transaction costs)

This backtest evaluates the portfolio generated by the regime-based allocation rule over the **full historical sample**, from 2006 and onward:

Metric	Strategy	SPY
CAGR	10.98%	10.81%
Annualized volatility	16.90%	18.57%
Sharpe ratio	0.557	0.562
Sortino ratio	0.792	0.797
Maximum drawdown	-27.82%	-55.19%
Annualized turnover	23.703	–

Table 7: Perf comparison between the regime-based strategy and SPY.

The regime-based strategy is compared with a buy-and-hold investment in SPY again. Without transaction costs, the strategy achieves a CAGR of 10.98%, slightly above SPY's 10.81%, it also exhibits lower annualized volatility, at 16.90% compared to 18.57% for SPY. Furthermore, the strategy also improves on a risk-adjusted basis. Its Sharpe ratio is 0.557, below SPY's 0.562. However, its Sortino ratio is slighlty lower: 0.792 vs 0.797.

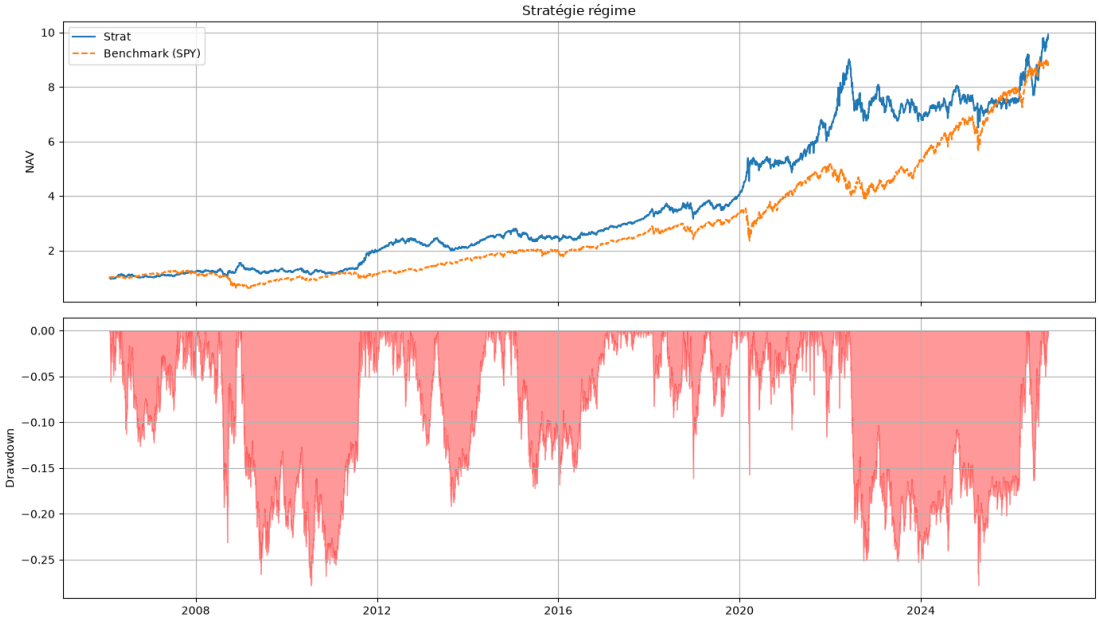


Figure 7: Strategy vs SPY (no tc) over full sample

The main advantage of the strategy is its drawdown profile. Its maximum drawdown is limited to -27.82%, substantially lower than SPY's -55.19%. The strategy therefore provides a similar long-term return with lower volatility and significantly improved downside protection.

Nevertheless, this protection is obtained at the cost of relatively high portfolio activity, with an annualized turnover of 23.703.

4.4 Historical Results and Comparison with S&P500 (including transaction costs)

The results presented above are theoretical, as they are based on gross returns and do not account for the frictions faced by a real investor. In practice, portfolio rebalancing involves transaction costs, bid-ask spreads, slippage, commissions, and potentially market-impact costs. These frictions are particularly relevant for strategies with high turnover.

To obtain a more realistic estimate of historical net performance, the analysis is processed again by imposing a transaction cost of 10 basis points on the portfolio turnover at each rebalancing date.

Metric	Strategy	Strategy (no tc)	SPY
CAGR	8.38%	10.98%	10.81%
Annualized volatility	16.90%	16.90%	18.57%
Sharpe ratio	0.468	0.557	0.562
Sortino ratio	0.573	0.792	0.797
Maximum drawdown	-0.3287	-27.82%	-55.19%
Annualized turnover	23.703	23.703	–

Table 8: Perf comparison between the strategy (with tc) and SPY.

The results deteriorate materially after transaction costs are introduced. Given the strategy's high annualized turnover, even a relatively small cost of 10 bps per unit of turnover generates a substantial drag on performance. The CAGR decreases, while the risk-adjusted performance and downside profile also worsen.

After transaction costs, the strategy no longer outperforms the passive S&P 500 benchmark in terms of overall performance. This result highlights that the apparent gross performance is not sufficient to establish economic viability. Reducing turnover and improving signal persistence are therefore essential areas for future development.

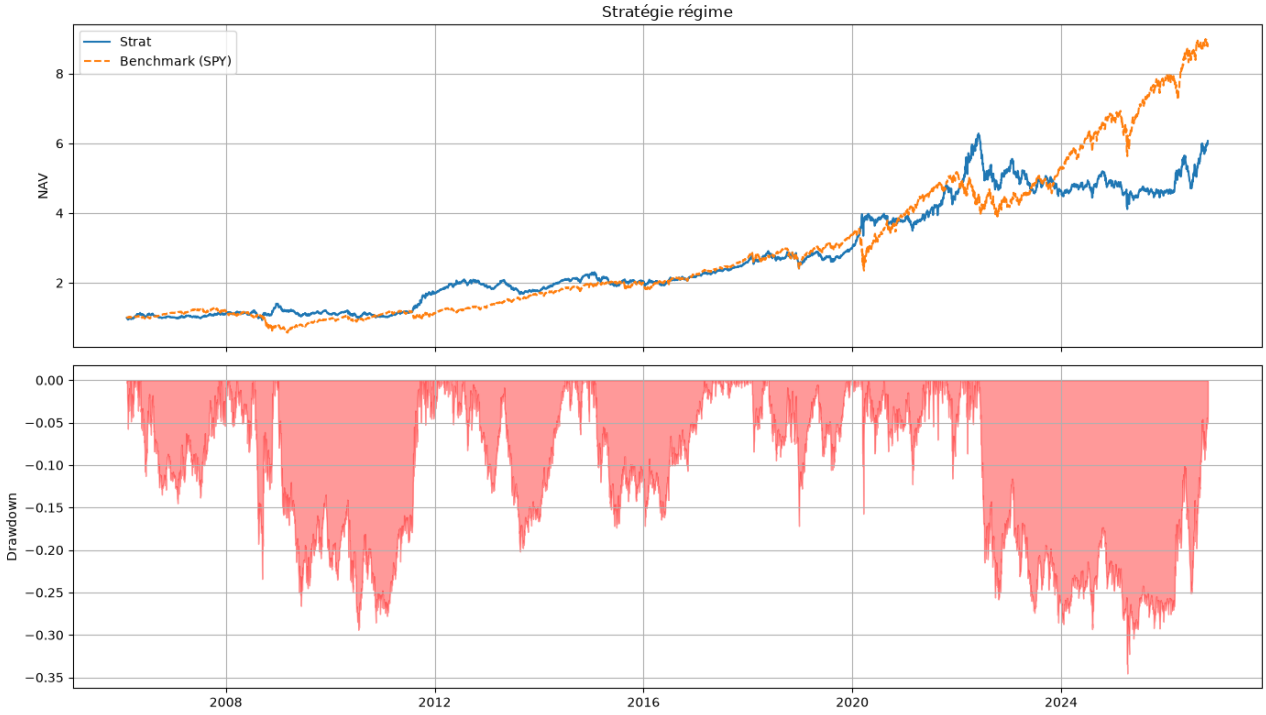


Figure 8: Strategy vs SPY (tc = 10bps) over full sample

5 Conclusion

The first version of the project provides a coherent and interpretable macro regime framework linking economic conditions to asset allocation decisions.

In gross terms, the strategy delivers performance close to SPY while significantly reducing maximum drawdown, which suggests that the regime engine captures economically meaningful differences across market environments.

However, the current allocation rule remains too simplistic, as it relies on binary 100% switches between single assets and generates high turnover.

Once transaction costs are introduced, the strategy no longer outperforms SPY. Therefore, at this stage, the model should not be viewed as an investable strategy in its current form. It should rather be interpreted as a **promising macro regime-classification prototype** that provides a solid base for more robust portfolio-construction and implementation work.

6 Areas for Improvement

6.1 Macro Model

- The model should be extended to include additional macroeconomic variables in the construction of the economic and policy scores.
- The current equal-weighting assumption should be reassessed. Future versions should investigate whether certain variables contain more relevant information than others and should assign differentiated weights accordingly.

6.2 Portfolio Allocation

- The portfolio-construction layer must be improved, as the current all-in, single-asset allocation rule is too rigid and costly to implement in practice.
- Cash positions should be introduced during periods of uncertainty or when the macroeconomic signal is weak, instead of maintaining a fully invested position at all times.
- The framework should evolve from binary regime selection toward diversified, regime-specific portfolio weights across several assets.
- The previously conducted correlation and copula analyses should be incorporated into the allocation framework to assess whether short positions could improve portfolio performance or risk management under specific market conditions.

6.3 Trading and Risk Management

- A broader risk-management framework should be introduced, including drawdown controls, stop-loss rules, volatility targeting, and explicit position-size constraints.
- These risk controls should be evaluated not only in terms of return enhancement, but also with respect to their impact on portfolio survivability, turnover, and implementation realism.
- Rules should be introduced to prevent excessive rebalancing when the regime changes frequently over short periods.
- Possible solutions include regime-persistence filters, minimum holding periods, signal buffers around regime boundaries, and turnover constraints.

"You should have a strategic asset allocation mix that assumes that you don't know what the future is going to hold. — Ray Dalio"

APPENDIX

Momentum score Construction

Let x_t denote the value of a given macroeconomic time series at time t. The function first converts the selected frequency into the corresponding number of observations per year:

$$k(f) = \begin{cases} 21, & \text{if } f = \text{daily}, \\ 1, & \text{if } f = \text{monthly}, \\ 3, & \text{if } f = \text{quarterly}. \end{cases} \tag{11}$$

For a selected lag L, the effective lag expressed in observations is defined as:

$$h = L\,k(f). \tag{12}$$

The raw momentum signal is then calculated as the change in the variable over the effective lag:

$$\Delta_h x_t = x_t - x_{t-h}. \tag{13}$$

For example, when the series is monthly and the lag is equal to six, the effective lag is six observations:

$$h = 6 \times 1 = 6, \tag{14}$$

and the raw momentum signal becomes:

$$\Delta_6 x_t = x_t - x_{t-6}. \tag{15}$$

The historical reference sample starts on 1 January 1985. All observations before this date are excluded from the estimation of the mean and standard deviation. Missing observations are also removed from the reference sample.

Let $\mathcal{R}$ denote the set of non-missing momentum observations from the reference period:

$$\mathcal{R} = \{\Delta_h x_t : t \geq t_0,\ \Delta_h x_t \text{ is observed}\}, \qquad t_0 = \text{1985-01-01}. \tag{16}$$

The mean of the reference momentum series is defined as:

$$\mu_\Delta = \frac{1}{|\mathcal{R}|} \sum_{\Delta_h x_j \in \mathcal{R}} \Delta_h x_j. \tag{17}$$

The standard deviation is computed using the population standard deviation:

$$\sigma_\Delta = \sqrt{\frac{1}{|\mathcal{R}|} \sum_{\Delta_h x_j \in \mathcal{R}} (\Delta_h x_j - \mu_\Delta)^2}. \tag{18}$$

The raw momentum signal is standardized using this reference mean and standard deviation:

$$z_t^{\text{mom}} = \frac{\Delta_h x_t - \mu_\Delta}{\sigma_\Delta}. \tag{19}$$

Finally, the standardized momentum score is winsorized by clipping it to the interval $[-C, C]$, where $C = 2.5$

$$m_t(x) = \min\{C,\, \max\{-C,\, z_t^{\text{mom}}\}\}, \qquad C = 2.5. \tag{20}$$

This clipping procedure limits the influence of extreme observations while preserving the direction of the signal. The resulting momentum score is therefore bounded between -2.5 and 2.5.

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Macro-Systematic Approach to multi-asset investing

Nicolas Poncin is a M.Sc. Financial Engineering student with a strong interest in global macroeconomics and multi-asset portfolio management. He aims to apply his academic background to real investment decision-making. This project was developed as a practical research initiative to explore how macroeconomic regimes influence asset behaviour and portfolio allocation, intending to build a disciplined and testable investment process linking macro analysis, portfolio construction and risk management. Before focusing on finance, Nicolas was a high-performance fencing athlete, representing his country at European and World Championships.



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